



NITI Aayog

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SECTOR AT A GLANCE

Biotechnology (S&T) and Industrial Research

Health Sector in India

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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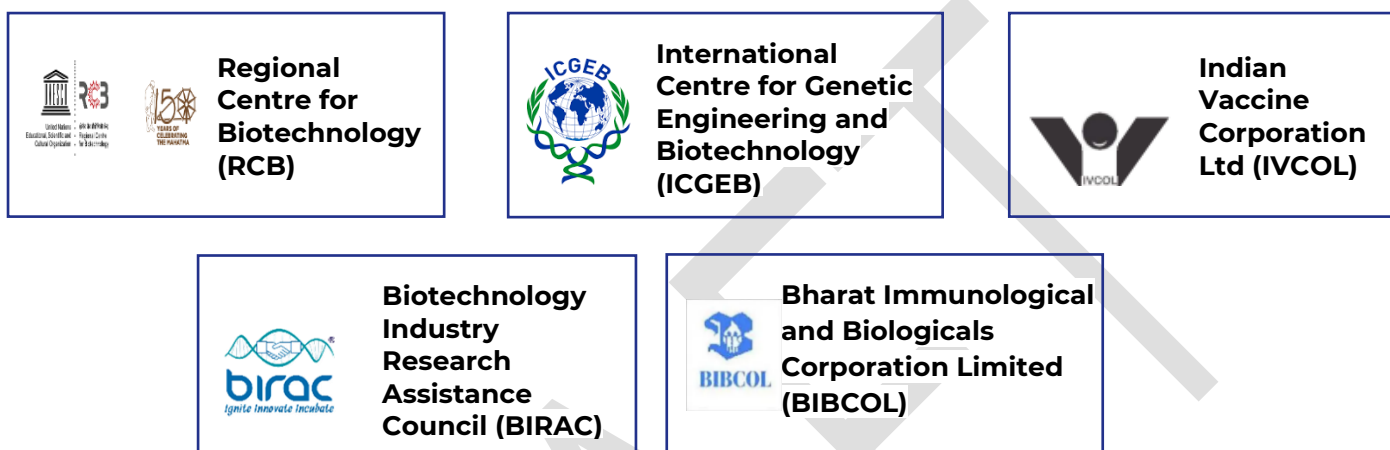
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Overview

Biotechnology is one of the sunrise sectors that is driving growth of the economy in India¹. It focuses on fostering cutting edge-research and innovation, with strong emphasis on translation research. The Department of Biotechnology (DBT), under the Ministry of Science and Technology, aims to make India a global leader in biotechnology by supporting research, innovation, and entrepreneurship to achieve National Development Programmes (NDPs) and Sustainable Development Goals (SDGs).

DBT works in collaboration with organisations to promote research activities for the growth of biotechnology industry.² These efforts have had a substantial impact on various sectors in India, including agriculture, healthcare, industry, and the environment. Below is the list of **international organisations and public sector undertakings** working in collaboration with DBT:



Vision²

Attaining new heights in biotechnology research, shaping biotechnology into a premier precision tool of the future for creation of wealth and ensuring social justice – specially for the welfare of the poor.



Mission²

Biotechnology is a cutting-edge field of science with vast potential to benefit humanity. The Department of Biotechnology is committed to achieving excellence in the promotion of biotechnology in India, in accordance with the Allocation of Business Rules, 1961, as enumerated in the Government of India notifications No. CD-172/86 dated February 27, 1986, and No. CD-87/87 dated January 31st, 1987. The department is tasked with delivering a range of services encompassing research, infrastructure development, generation of human resource, promotion of industries, popularization of biotechnology, establishment of centres of excellence, enforcement of biosafety guidelines for genetically modified organisms and recombinant DNA products, as well as the implementation of biotechnology-driven programmes to benefit society. A significant undertaking within this mission is the establishment of a comprehensive bioinformatics system that aims to create a global information network for the scientific community. The organisations mission is:

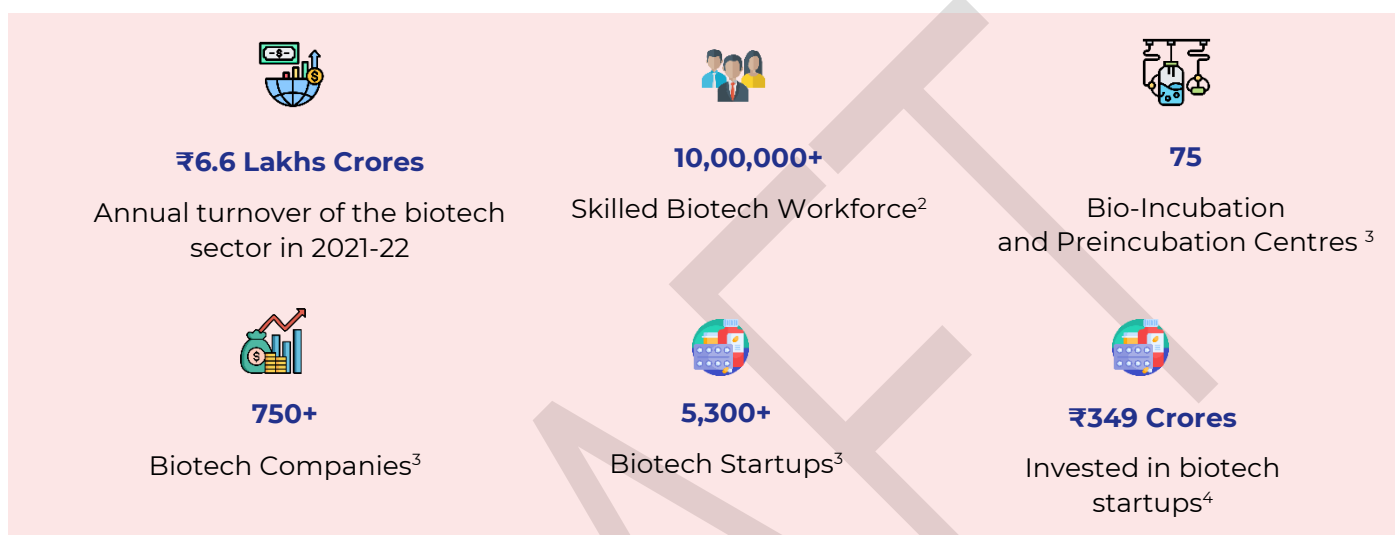
- Realising and promote full potential of biotechnology industry.
- A strategic and substantial investment to develop products, processes, and technologies.
- Improve the effectiveness, productivity, and cost-efficiency of molecular medicine, agriculture, nutritional security, enhance the knowledge base, empower the scientific and technological capabilities of human resource, environmentally sustainable technologies, nurturing the leads of potential utility, a strong infrastructure for research and commercialization, and bringing the bioproducts to the market place.
- Socio-economic development / applicants of biotech for upliftment of SC/ST, women, and rural population.

1. Department of Biotechnology India (2021). *National Biotechnology Development Strategy (2021-25)*.

2. Department of Biotechnology (2022). *Annual report 2022-23*.

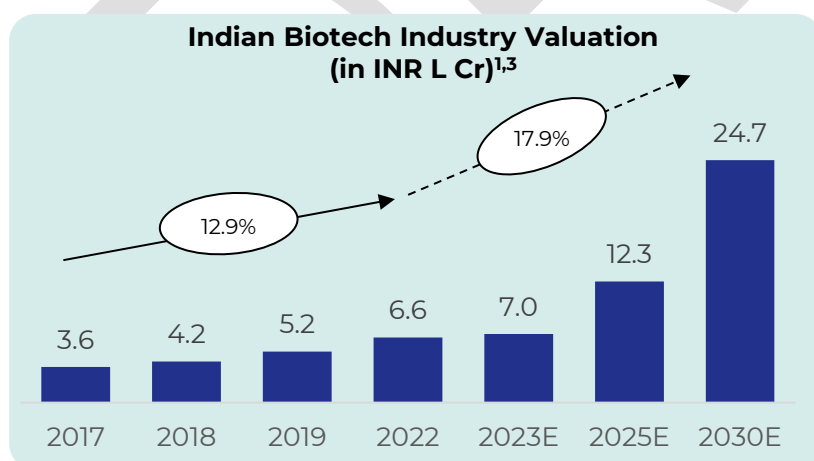
Biotechnology Landscape in India

India is ranked **12th in the world** and **3rd in Asia-Pacific** region in the biotechnology sector³. India contributes **3% share** in the global biotechnology industry and ranks **46th in the global Innovation Index**². The government is promoting biotechnology innovation missions and entrepreneurship through the Aatmanirbhar Bharat initiative, leveraging strategic partnerships and building capacities across the country. Despite the COVID-19 pandemic, which caused two rounds of lockdowns and other disruptions, 2021 was a remarkable year for India's bioeconomy. This is because the biotechnology industry played a leading role in fighting the pandemic by developing and deploying vaccines, antivirals, diagnostic tests, and so on.



Indian Biotechnology Industry is valued at **INR 6.6 Lakhs Crores (2022)** and is expected to grow at a **CAGR of 17.9%** between 2022-30.³ According to "India Bioeconomy Report 2022" released by DBT, the Biotech industry tripled its R&D spending to **INR 8,460 Crores in 2021 from INR 2,986 Crores in 2020** driven by the development of COVID-19 products and solutions.³ The goal is to support and nurture the bio economy to achieve the anticipated **INR 24.7 Lakhs Crores by 2030.**²

Figure 1 - Biotechnology Industry Valuation (2017-2030)



Indian Biotechnology Industry is expected to grow at a **CAGR of 17.9%** between 2022-30³

Biopharma has the majority share (**around 50%**) in the overall Biotechnology market in India⁴

3. Invest India (2023). Biotechnology. Retrieved 19 September 2023

4. BIRAC (2022). India Bioeconomy Report 2022

Biotech Companies in India

India witnessed a significant growth in the number of **biotech startups** that increased to **5,365 in 2022 from 4,237 in 2020** owing to the growing enabling ecosystem. The cumulative start-ups grew by 26.6% in 2021 (term “cumulative” stated means the total start-ups). For the first time in India, the biotech industry has witnessed a remarkable achievement with the registration of more than 1,128 new start-ups in a single year, compared to 840 new start-ups in 2020. The Government aims to increase the Biotech Start-ups arising from strong talent pool 2 times, to 10,000+ by 2025. India aims to be **recognized as a Global Bio-manufacturing Hub by 2025** and figure among the top 5 countries of the world.⁴

This impressive growth of start-ups can be attributed to the vision and dedicated efforts of the DBT in fostering innovation and R&D. Some of the biotech companies in India are *Biocon, Biological E, Serum Institute of India, Advanced Enzymes, Bharat Biotech, MetaHelix, Strand Life Sciences, Shantha Biotech, and xCyton Diagnostics* etc.⁴

Table 1 - Some Prominent Manufacturers in the Indian Biotechnology Sector

Biosimilars ⁵	Vaccines ⁶
Reliance Life Sciences Group	Bharat Biotech International Ltd
USV Ltd	Biological E Limited
Inbiopro Solutions	Serum Institute of India
BioGenomics	Shantha Biotechnics Ltd.
Biocon Ltd	Cadila Healthcare
Zydus Cadila Group	Ranbaxy Lab
Serum Institute of India	GSK Asia Pvt Ltd
Enzene Biosciences Ltd	Hetero Biopharma Limited
Dr. Reddy's Laboratories Ltd	Indian Immunological Limited
Chandra Bhagat Pharma Pvt Ltd	Chiron Behring
Lupin Ltd	Dano Vaccine and Biological Pvt. Ltd
Cipla Ltd	Green signal BioPharma Ltd
RAS Lifesciences Pvt Ltd	Sanofi Pasteur India Pvt Ltd
Neiss Labs Ltd	Haffkine
Intas Biopharmaceuticals	Pasteur institute of India

Maharashtra, Karnataka, and Delhi (NCR) serve as the biggest hubs for Biotech Startups in India.³ Companies focused on vaccine manufacturing such as Serum institute of India, Biological E, Bharat Biotech, Zydus Cadila, Aurobindo Pharma, Stelis BioPharma, Shilpa Biologicals, Dr. Reddy's Lab etc. has increased their installation capacities (by million doses) in 2021.⁴ Some of foreign companies that invest in India are Bosch (Germany), Limagrain (France), Tekes (Finland), Mylan (US), BPI France (France), GE Healthcare (US), Abbott Laboratories (US)

⁵ Birac (2019). India: The emerging hub for biologics and biosimilars

⁶ India Science, Technology & Innovation (n.d.). *List of Licensed Human Vaccine Manufacturing Facilities in India*. Retrieved on 19 September 2020

Figure 2 - Major Regional Bases of Biotech companies in India (2021)⁴

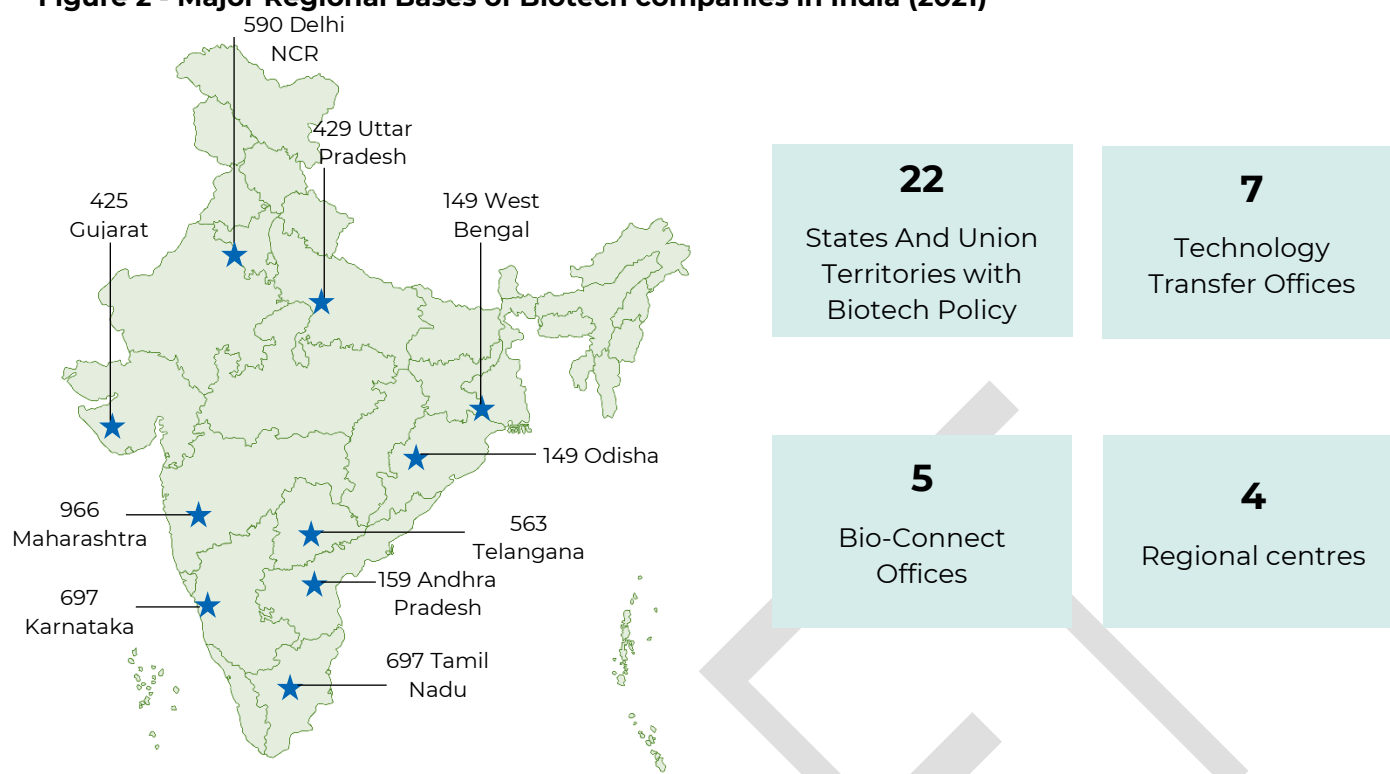


Figure 2: Regional bases of biotech companies in India

DBT has taken various initiatives in these states to boost the biotechnology industry of India. As part of the DBT UMMID (Unique Methods of Management of Inherited Disorders) initiative, Indian Council of Medical Research (ICMR) has set up facilities for antenatal screening of hemoglobinopathies and newborn screening for six inherited diseases in the **Aspirational District** of Nandurbar, Maharashtra.⁷

7. Indian Council of Medical Research, ICMR (2021). Most Important Outcomes (2021-22)

7a. Ministry of External Affairs.(2020). India BIRAC Initiative.

7b. Department of Biotechnology. (2016). BIRAC success story.

7c. Department of Biotechnology. (2022). BIRAC Annual Report.

Research Landscape in Biotechnology

Government of India is committed to promote biotech-based drug development under the **"Make in India"** campaign. Multiple reports have summarized initiatives introduced by the DBT, Biotechnology Industry Research Assistance Council (BIRAC) and the government of India to support the vision of making India a hub for biotechnology-based innovation and research such as National Biotechnology Development Strategy, other new initiatives include Project Development Cell (PDC), Technology clusters, and Investment Clearance Cell (ICC).⁵

BIRAC supports innovative research that can be translated into commercial products. It encourages public-private partnerships to attract investment from industry, investors, and other agencies. BIRAC also provides funding to startups, from INR 50 Lakhs to INR Seven Crores, to support them develop their ideas. The Biotechnology Innovation Fund – AcE (Accelerating Entrepreneurs) has mobilized more than INR 300 Crores to support biotech startups in scaling up their Research and Development and innovation. In addition to startups and biotech companies, BIRAC also supports research institutes and small and medium-sized enterprises (SMEs). Over 1,000 entrepreneurs and startups have been supported by BIRAC, and over 130 products and technologies developed with their support are now available in India and other countries.⁵

BIRAC has launched and spearheaded many national programmes such as BIG (Biotechnology Ignition Grant), SBIRI (Small Business Industry Research Initiative), BIPP (Biotechnology Industry Partnership Programme), and SPARSH (Social Innovation Programme for Products Affordable and Relevant to Societal Health) that spur the Indian biotech enterprise to drive innovative R&D for productization from testing their ideas to POC as well as taking it further the product development funnel.^{7a}

BIRAC has supported many successful products and technologies. For instance, the Hyderabad based LV Prasad Eye Institute, in partnership with Sheffield University, developed and used new biocompatible material from stem-cell-based therapy to restore eyesight. Cadila Pharmaceuticals Ltd.'s combination pill to control cardiovascular disease. AIIMS is working on a preliminary prototype of a chest compression device for cardiac arrest patients, is another example of successful products.^{7b}

In 2021-22, BIRAC took up a few new initiatives including setting up of BioNEST incubator clusters, launch of focused calls, partnering with private organisations such as NASSCOM (National Association of Software and Service Companies), GCI (Grand Challenges India), IKP knowledge park.^{7c}

DBT has funded various projects in different areas including biomedical engineering, vaccine research, stem cell and regenerative medicine research, oncology research, drug research, biotech parks, and human genomics.³



Vaccine: DBT supports vaccine R&D in areas such as candidate vaccine development, pre-clinical and clinical vaccine development, and human immunology of infectious diseases. The vaccine component of the mission focuses on accelerating the development of vaccines for complex, emerging, and high-priority infections in India. Currently, 13 vaccine candidates for COVID-19, flu, pneumonia, cholera, and dengue which are at different stages of development, are being supported by the mission.



Oncology research: DBT provide resources to investigators and institutions to carry out basic, applied, translational and clinical research. This research has led to new insights into the causes and treatment of cancer, including investigation on hormone therapy sensitive and resistant breast cancer using multiomics*, big data generation approaches, study of clonal evolution of patient tumours in response to therapy over time using patient samples before drug administration and exploring novel inhibitors B-cell lymphoma 2 (BCL2)-specific inhibits against leukaemia and lymphoma.



Stem Cells and Regenerative Medicine: DBT is supporting basic biology research, early and late translational research, developing gene editing technology for possible therapeutic applications, creation of animal models for various human diseases.



Genome Editing Applications in Healthcare: DBT has made efforts to encourage R&D programmes in emerging genome engineering technologies, such as CRISPR-Cas9 (Clustered Regularly Interspaced Short Palindromic Repeats) and development of new tools and technologies for efficient delivery of genome editing constructs, such as viral vectors and nanoparticles.



Biomedical Engineering: DBT supports innovative ideas related to development of technology for manufacturing microneedles (MNs), standardization of fluorescence-based probes selective for Human Serum Albumin (HSA) and creatinine detection, and development of HDPE* and UHMWPE* based biomaterial.

*Multiomics: A new approach where the data sets of different omics groups are combined during analysis. The different omics strategies employed during multiomics are genome, proteome, transcriptome, epigenome, and microbiome

*HDPE: High-Density Polyethylene

*UHMWPE: Ultra-High-Molecular-Weight-Polyethylene

Trends in Biotechnology

Through research and development, DBT is addressing key challenges in medical biotechnology. Technological interventions are being made to be future ready through the development of affordable vaccines, gene editing products, and diagnostics.

The sub-sectors of biotechnology are rapidly evolving, with new trends emerging all the time. Some of the most promising sub-sectors include medical interventions, human genomics and oncology, computational biology, energy and environment, biotech parks and clusters, and biotech R&D. These sub-sectors are growing due to a number of factors, including:

- **Robust research and development:** There are a lot of investment in biotechnology research, which is leading to new discoveries and innovations.
- **Increasing number of start-ups:** There are a growing number of startups in the biotechnology industry, which are bringing new ideas and technologies to market.
- **Rising disease burden:** The prevalence of chronic diseases is increasing, which is driving demand for new treatments.

Sub-sectors of Biotechnology



Medical Interventions

Medical Intervention focuses on development of new drugs particularly in the areas of vaccines, stem cell therapy, and biomedical engineering research.



Human Genomics and Oncology

Human genomics and oncology include personalized drugs, DNA sequencing, genome-panel assays, and genome-wide microarrays.



Computational Biology

Computational biology focuses on development of Artificial Intelligence (AI) tools to advance diagnosis & treatment.



Energy and Environment

Energy and environment include research in the field of biofuels and facilitate the transition of processes and technologies from research to large-scale implementation and commercialization.



Biotech Parks and Clusters

Biotechnology Parks are developed to support the essential **infrastructure** for the conversion of research into products and technology.



Infectious Disease Biology

Basic and applied research for better care for infectious diseases such as HIV/AIDS, tuberculosis (TB), vector borne diseases; influenza, Japanese Encephalitis (JE) and neglected tropical diseases.

A. Medical Interventions

1. Biomedical Engineering

Biomedical Engineering represents a profoundly interdisciplinary area of research that draw upon knowledge from engineering and various quantitative sciences to unravel the intricacies of biological systems. It offers cost-effective solutions aimed at enhancing the overall quality of life. With a primary focus on studying emerging diseases, genome manipulation and engineering, the development of transgenic systems and creation of genome-based diagnostics, DBT plays a crucial role in advancing the field of biomedical engineering. Its contribution is directed toward the development of tissue engineering, diagnostics, devices, biosensors, bio composites, etc.



Major Projects²

UMMID initiative (Unique Methods of Management and treatment of Inherited Disorders); ATGC (Accelerated Translational Grant for Commercialization) supports research, development, and implementation of innovative diagnostic methods and treatments for inherited genetic disorders, aiming to enhance the management and quality of life for affected individuals.⁸

ATGC Scheme aims to enable academic researchers to accelerate the translation of research outcomes into viable products and bridge the gap between academic research and commercialization in the biotech industry.⁹

2. Vaccine Research

Vaccines stand out as among the most economically efficient healthcare interventions in the fight against infectious diseases. Consequently, for the past three decades, the DBT has been dedicated to enhancing the research and development landscape for vaccines in India. This commitment is evident in their pioneering strategies, such as cultivating collaborative partnerships with prominent global experts and implementing inventive funding mechanisms.

Noteworthy international partnerships in the field of vaccine, have contributed immensely to the science of vaccinology, encompass initiatives such as the Indo-US Vaccine Action Programme (VAP); bilateral co-operations with countries like Australia, Finland, Norway, and France; strategic engagement with the Coalition for Epidemic Preparedness Innovations (CEPI).

India Supplies Vaccines to more than **150 countries** in the world¹⁰

Development of India's first indigenous quadrivalent human papillomavirus (f) vaccine, known as Cervavac²

Monthly production capacity of 20 million doses for COVAXIN after facility enhancement²

8. Department of Biotechnology. (n.d.). *Computational Biology: UMMID Initiative*. Retrieved on 19 September 2023

9. Birac (2023). *Guidelines for Projects Supported under DBT-BIRAC Accelerated Translational Grant for Commercialization (ATGC) Scheme*

10. Pharmaceuticals (2023). *Annual Report 2022-23*

India holds a prominent position as a global leader in the provision of DPT, BCG, and Measles vaccines. Moreover, India stands as one of the major contributors to the worldwide distribution of affordable vaccines. Every year, India provides 150 Crores vaccine doses, exporting them to more than 150 countries and generating an estimated INR 3,363 Crores in global sales¹¹



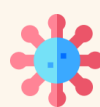
INR 252 billion

is the projected valuation of Indian vaccine market by 2025¹²



40-70%

of WHO's demand for DPT, BCG vaccines used to combat TB is fulfilled by India¹⁰



90%

of WHO's demand for the measles vaccine is covered by India¹⁰



>220 Crores

COVID-19 Vaccination doses distributed¹³

(as of 4 Sep 2023)

Immunization¹⁴

The Universal Immunization Programme (UIP) of India is one of the largest public health initiatives, with an annual target population of 2.7 Crores new-borns and 2.9 Crores pregnant women.

Mission Indradhanush launched in December 2014, has the aim of full immunization coverage for 90% of children while focusing on hard-to-reach areas of rural India.

Vaccines launched during CoVID-19



COVAXIN®, Bharat Biotech's indigenous COVID-19 vaccine, was created in collaboration with the Indian Council of Medical Research (ICMR) - National Institute of Virology (NIV).¹⁵



It is the largest producer and seller of vaccines in the world and are used in 170 countries for their immunization's programmes. It has crossed many milestones in technological advancement by installing robotic arm for virus handling, cell factory, facilitating manufacture of recombinant vaccines and monoclonal antibodies along with COVID 19 Vaccine (**Covishield**).¹⁶



Zydus Cadilla is mostly centred in discovering and developing Novel Biologicals, Biosimilars and Vaccines. During CoVID-19 pandemic, they successfully received approval for **ZyCoV-D** for Emergency Use Authorization (EUA) from the Drug Controller General of India (DCGI).⁶



Dr. Reddy's Laboratories Limited and Sputnik LLC jointly conducted multi-centre, phase II/III adaptive clinical trial to assess safety and immunogenicity of Gam-COVID-Vac combined vector vaccine (**Sputnik**) during pandemic.⁶



Mission COVID Suraksha, a Indian COVID-19 vaccine development Mission supported nation's first protein subunit vaccine for COVID-19, **Corbevax™** developed by Biological E, that received Emergency Use Authorisation (EUA) for the 12-18 years age group on February 21, 2022.⁶

11. NITI Aayog (2021). *Investment Opportunities in India's Healthcare Sector*

12. Indbiz (2020) *Ancillary firms look to tap US\$6 billion vaccine business*. Retrieved 20 September 2023

13. MyGov (2023). *Covid-19*. Retrieved 20 September 2023

14. National Health Mission (2023). *Immunization*

15. Bharat Biotech (n.d.). Retrieved 19 September 2023

16. Serum Institute of India (n.d.). Retrieved 19 September 2023

Table 2 - List of licensed human vaccines manufactured in India⁶

Type of Manufacturer	Manufacturing company	Licensed vaccines
Indigenous Manufacturers In Government	BCG vaccine, Guindy, Chennai.	BCG, Tuberculine
	CRI, Kasauli	DTP, yellow fever, JE, TT, DT, Concentrated DTP vaccine
	Pasteur institute of India	DTP, TT, DT & inactivated Rabies vaccine
Indigenous Manufacturers in Public Sector Undertaking	BIBCOL, U.P.	bOPV
	Haffkine	mOPV & bOPV
	Human biological institute	Rabies, DTP, TT, DT & Hep- B
	HLL Biotech Ltd.	Hep B, DTwP- HepB-Hib
Indigenous Manufacturers in the Private Sector	Bharat biotech International Ltd	Hib, Rabies, bOPV, mOPV, DTP+Hib+HepB, Vi polysaccharide Typhoid, H1N1, DTP, DTP+HepB, Rotavirus vaccine, Inactivated JE vaccine, Typhoid+TT Conjugate Vaccine & DTP+Hep- B+Hib (Liquid), DTP+Hib
	Biological E	DTP, TT, JE bulk & DT, Diphtheria, Pertussis, DTP+Hep-B, Hib, Heb-B, IPV, HepB+Hib (Liquid & Lyophilized),
	Biomed Pvt. Ltd	Hib, Meningococcal Polysaccharide (A,C, Y,W 135), bOPV, Vi Polysaccharide Typhoid Vaccine & Meningococcal polysaccharide (A & C), Rabies
	Zydus Lifesciences (formerly known as Cadila healthcare)	Rabies vaccine, Bulk Rabies vaccine, Pandemic influenza H1N1 2009 monovalent vaccine, Typhoid Vaccine, Rabies vaccine, Tetravalent Influenza (split virion), H1N1 VLPs Vaccine, TT, BCG
	Cadila Pharmaceuticals	H1N1 VLPs Vaccine
	Chiron Behring	Rabies vaccine
	Dano Vaccine & Biological Pvt. Ltd.	TT
	Green signal BioPharma Ltd.	BCG Vaccine
	Panacea	DTP, HepB, DTP+Hep-B, DPT+Hep-B+Hib, IPV, Hib & H1N1, bOPV, DTP-Hib, DPT+Hep-B+Hib+IPV, wP
	Ranbaxy Lab	Typhoid polysaccharide & Hib Conjugate vaccine
	Serum Institute of India	DTP, TT, DT, Hep-B, Hib (Vaccine & bulk), MMR, Measles, Rubella, BCG, Rabies, IPV, DTP+Hep B+Hib (Liquid +lyophilized), DTP+HepB, DTP+Hib, H1N1, Meningococcal A conjugate (Freeze dried), Mumps, MR, H1N1(whole virion inactivated), Measles +mumps,

Indigenous Manufacturers in the Private Sector		Measles+ Rubella, Influenza, Vaccine seasonal, Diphtheria Vaccine (bulk), TT bulk, Pertussis bulk, Measles bulk, Mumps bulk, Rubella bulk & DT bulk & OPV vaccine, CRM 197 Bulk, DTP+ Hep B+ Hib Bulk
	Shantha Biotechnics Ltd.	DTP, DTP+HepB+Hib (Liquid), DTP+Hib, DPT+Hep B, TT, Hib, Hep-B, DT bulk, TT Bulk, Hib Bulk, Hep B Bulk, DTP bulk, DTP+HepB+Hib bulk, DTP+HepB+Hib RTF bulk, Oral cholera vaccine, IPV RTF Bulk, IPV, Inact. B. pertussis bulk
	GSK Asia Pvt. Ltd	Pneumococcal Polysaccharide and Non-Typeable Haemophilus influenza vaccine (packing and labeling)
	Sanofi Pasteur India Pvt Ltd	Hib, DTP-Hib, Typhoid, Hep A, Pneumococcal, Yellow fever, Seasonal influenza, Rabies, Meningococcal, IPV, DTaP-IPV-Hib, Varicella, DTaP

Major Projects²

The **Ind-CEPI (Coalition for Epidemic Preparedness Innovations) Mission** aims to develop vaccines for high-transmission diseases (Zika, Coronavirus, Nipah, Monkey fever, Chikungunya, and others) in India, while enhancing public health system coordination and vaccine industry collaboration to address current and future threats.²

The **Indo-US Vaccine Action Programme (VAP)** is a programme jointly implemented by Indian and US bodies to support the research and development of new and innovative vaccines and other related activities.²

Other Initiatives:

Vaccine Maitri: India's friendship initiative

India launched the Vaccine Maitri (Vaccine Friendship) initiative - a major diplomatic effort to gift and supply made-in-India Covid-19 vaccines to low-income and developing countries globally following its philosophy of "Vasudaiva Kutumbakam".¹³

A total of **28.1 Crores** doses of Made-in-India COVID-19 vaccines were dispatched to 99 countries along with UN Peacekeepers and UN Health Workers as part of the Vaccine Maitri initiative as of November 30th 2022.¹⁸

Table 3 - Top 5 COVID-19 vaccine supply across countries as of 2023

Country	Total Supplies in Lakhs (as of June 2023) ¹⁷
 Netherlands	1,136
 Australia	309
 Bangladesh	281
 Myanmar	212
 Nigeria	98

Highlights



Successful administration of two vaccines, namely Bharat Biotech's COVAXIN and Oxford-AstraZeneca's Covishield (manufactured by SII), played a crucial role in protecting the Indian population from the impact of COVID-19.¹⁰



The Government of India's Aarogya Setu, which was developed through Public Private Partnership to connect health services and the people of India, had augmented the Govt's initiatives in proactively reaching out to and informing users about the potential risk of infection, best practices, and relevant medical advisories pertaining to the containment of the COVID-19 pandemic.¹⁹



Indian government has introduced the Electronic Vaccine Intelligence Network (eVIN) system, which digitalizes the management of vaccine stock, logistics, and temperature tracking across all levels of vaccine storage, ranging from the national level down to the sub-district level.¹⁴



National Vaccine Policy of India focuses to strengthen various vaccination programmes to address the issues of vaccine security, vaccine research and development, vaccine quality regulatory system, and product development for public health emergency.²⁰

3.Stem Cells and Regenerative Medicines



Major Projects²¹

The researchers from the **Centre for Stem Cell Research (CSCR)**, CMC Vellore, have obtained approval from Central Drugs Standard Control Organisation (CDSCO) for Phase I clinical trials in India for haemophilia A gene therapy.

The **National Guidelines for Stem Cell Research** provides guidelines for individuals and institutions undertaking research involving any kind of human stem cells and their derivatives in India.

17. Ministry of External Affairs (2023). *Vaccine Supply*. Retrieved on 15 June 2023

18. MEA (2022). *Ministry of External Affairs Annual Report 2022*

19. India (n.d.). *Fight against COVID-19*. Retrieved 19 September 2023

20. Ministry of Health and Family Welfare (2011). *National Vaccine Policy*

21. Department of Biotechnology (2017). *National Guidelines for Stem Cell Research*

B. Human Genomics and Oncology

1. Human Genomics

The Human Genetics and Genomics programme under the DBT provides research and development support to unravel the impact of the genome on human health and disease. This programme encompasses a wide spectrum of research, including translational research, clinical research, and basic research, all aimed at understanding how genomic information can be applied to enhance medical practices.



Major Projects²²

The **Genome India Project** aims to gather 10,000 genetic samples from individuals in India to investigate genetic variations, disease prevalence, and personalized healthcare in India.

The **National Genomics Core** has been established by the DBT, Government of India which offers genomic services like DNA sequencing, genome-panel assays, genome-wide microarrays, etc. to all individuals and institutions across the country.

2. Oncology Research

Cancer is a complex disease. Many subtypes of the disease have distinct clinical features and susceptibilities to therapy. Many cancers are still undiagnosed until they enter at advanced stages. DBT has a clear role in the management and control of the disease. Through R&D programmes, multi-institutional networking programmes, DBT has provided resources to individual investigators and to institutions, to carry out research and develop new methods to prevent and treat disease and conduct research especially in challenging areas pertaining to different types of cancer.



22. Department of Biotechnology (n.d.). *Human Genomics Oncology - Cancer*. Retrieved 19 September 2023

Major Projects²²

Under a collaboration of **Virtual National Cancer Institute**, multi-omics data generation approaches are used to understand hormone resistance in Breast Cancer.

The **ICGC (International Cancer Genome Consortium) DBT India Project** studies genomic alterations in Indian cancer patients, aiming to improve diagnostics and treatments. India is a founding member of this consortium.

C. Computational Biology

1. Theoretical and Computational Biology

In 1987, DBT initiated the field of Bioinformatics as a key focus area in Biotechnology, providing continuous support that propelled the growth of Bioinformatics in India. This support included the establishment of essential infrastructure, including a **Supercomputer**. Over time, Bioinformatics has evolved globally, drawing contributions from diverse fields such as Physics, Mathematics, Pharmaceutical sector, and emerging biotechnology areas like Genomics and Proteomics. As research and development activities in these areas generate vast amounts of data, the scope of Bioinformatics has expanded significantly. Considering international advancements, DBT has broadened the scope of Bioinformatics and redesignated it as 'Theoretical and Computational Biology (TCB)'.

Bioinformatics has effectively utilized for various purposes such as exploring biological prospects, conserving and managing bioresources, assessing products, processes, and raw materials, handling complex data for planning and monitoring major national programmes, and meeting the growing demands for contract services and business outsourcing in the pharmaceutical and biotechnology sectors. A significant challenge in fully leveraging Bioinformatics to address life science issues lies in formulating appropriate computational biology problems that can be effectively tackled using IT tools. The solution lies in cultivating strong leadership with expertise spanning both life sciences and information technology, along with establishing robust institutional collaborations between specialists.



Major Projects²³

Imaging BioBank for Cancer is an ongoing project, to develop AI tools and database for advance research in cancer and also aiming cancer diagnosis/ prognosis and cancer care.

MANAV: Human Atlas Initiative project is helping to deliver an open-source annotation platform, organ (skin) model development, up-skilled manpower and community.

23. Department of Biotechnology (n.d.). *Computational Biology – Theoretical and Computational Biology*. Retrieved on 19 September 2023

D. Energy and Environment



Major Projects²

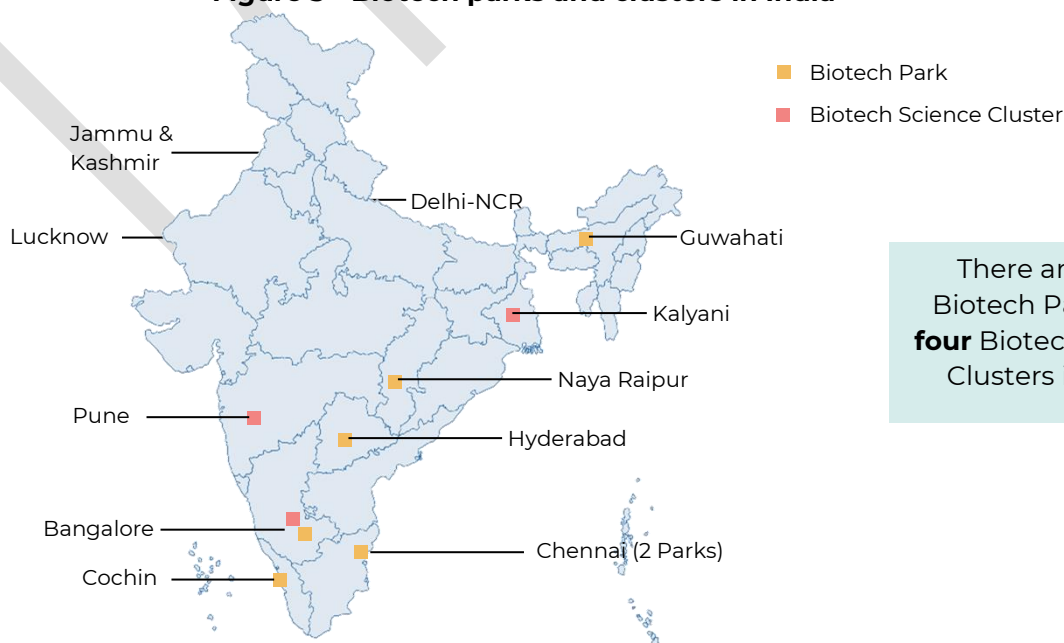
The establishment of five **Bioenergy Centres** by the department aims to enhance the research foundation in the field of biofuels and facilitate the transition of processes and technologies from research to large-scale implementation and commercialization.

Mission Innovation is a collaborative effort involving 23 countries and the European Commission with the goal of revitalizing and expediting global innovation in clean energy. The objective is to increase the accessibility and affordability of clean energy on a global scale.

E. Biotech Parks and Clusters

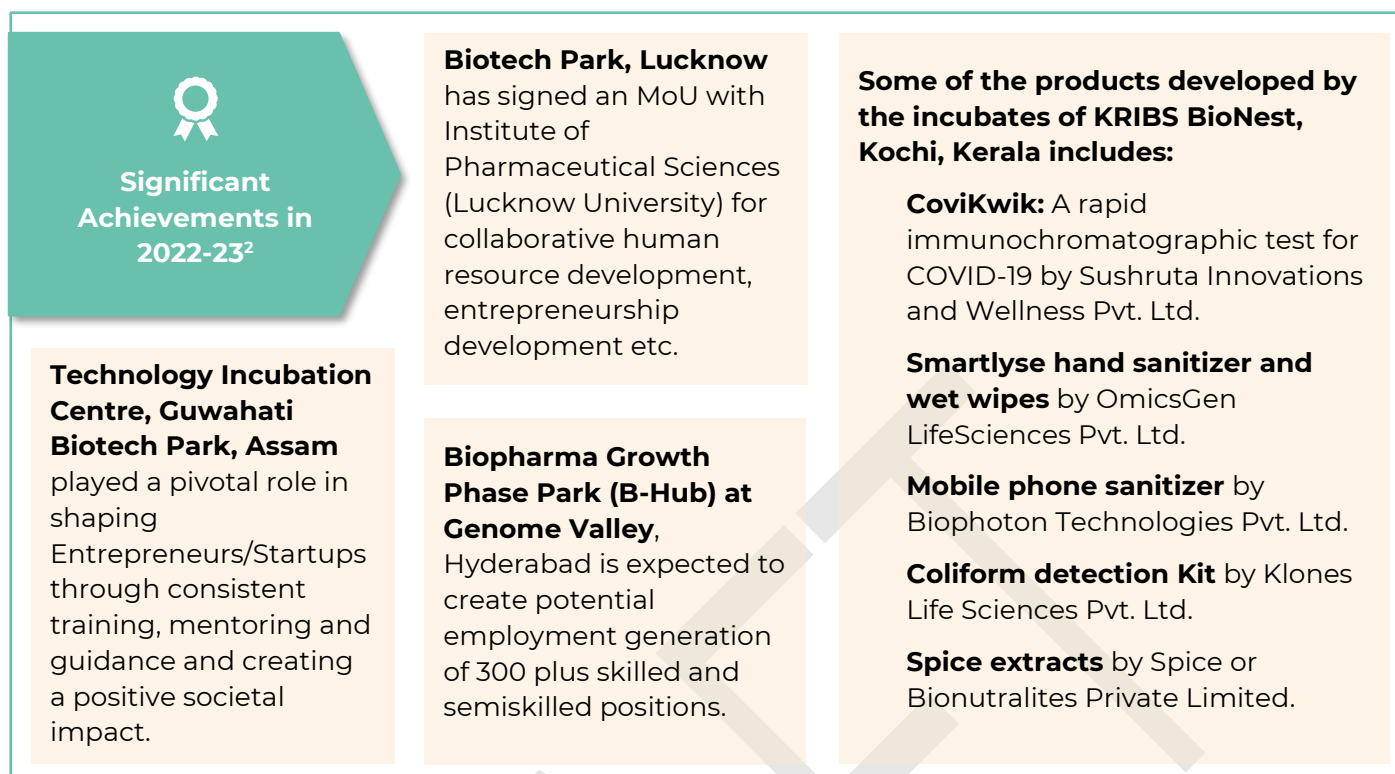
DBT has created Biotechnology Parks/Incubators nationwide with the aim of transforming research outcomes into products and services. These facilities offer essential infrastructure support and serve as platforms for scientists and Small and Medium-sized Enterprises (SMEs) to engage in technology incubation, demonstrate their innovations, and conduct pilot plant studies, all of which expedite the commercial development of the biotechnology sector in India. The aim of **National Biotechnology Parks Scheme** is to develop high quality basic infrastructure and high-end central equipment facilities with the required support services for undertaking Research and to boost biotechnology sector in the country. To date, the department has supported nine biotechnology parks located in various states, for driving commercialization of novel technologies, nurturing and sustaining emerging ventures, and facilitating meaningful collaborations between new enterprises and other stakeholders within the biotechnology sector, including academia and government.²⁴

Figure 3 - Biotech parks and clusters in India



There are **nine** Biotech Parks and **four** Biotech Science Clusters in India

24. Department of Biotechnology (n.d.). *Biotech Parks and Incubators*. Retrieved 19 September 2023



F. Infectious Disease Biology



1. Infectious Disease Biology-1 (IDB-1)

Infectious diseases rank among the top 10 leading causes of death in the country, with a predominant prevalence of conditions like tuberculosis, neonatal disorders, diarrhoeal diseases, and lower respiratory infections. India faces a considerably higher disease burden or Disability-Adjusted Life Years (DALYs), for conditions such as iron-deficiency anaemia, tuberculosis, and diarrheal diseases with rates 2.5 to 3.5 times greater than the global average.

Within the framework of Infectious Disease Biology (IDB) -1, focusing on **Bacterial and Fungal Diseases**, the Department is actively supporting both fundamental and applied research. These efforts aim to enhance understanding of infectious diseases caused by bacterial and fungal pathogens, with a specific focus on globally significant concerns like Tuberculosis, including Multi Drug-resistant (MDR) and Extensively Drug-Resistant (XDR) TB, as well as the growing challenge of Anti-Microbial Resistance (AMR).

Major Projects²

Dare2eraD TB

- The DBT initiated a programme called **Dare2eraD TB**, aimed at harnessing data-driven research to combat tuberculosis. This comprehensive TB programme encompasses several key initiatives, including organising Indian TB Knowledge Hub Webinar Series, setting up the Indian Tuberculosis Genomic Surveillance Consortium, developing host directed therapies against TB and treatment regimens for extra-pulmonary TB.
- Under the Indian TB Knowledge Hub Webinar Series, 8 webinars were organised until now.

Mission AMR

National Action Plan on Antimicrobial Resistance (NAP-AMR) had launched this mission and joined Global AMR R&D Hub; 10 projects have been supported for developing new AMR treatment options.

2. Infectious Disease Biology-2 (Disease biology of Viral and Protozoan diseases, HIV, Malaria)

The Infectious Disease Biology (IDB) Programme aims to provide solutions to infectious diseases of global concern such as HIV/AIDS, tuberculosis (including MDR and XDR TB), vector borne diseases; emerging or re-emerging threats such as influenza, Japanese Encephalitis and antibiotic-resistant microbes in terms of therapeutics, diagnostics and preventive measures. The Programme supports R&D projects with an aim to achieve the Sustainable Development Goals (SDG) Target 3.3 of ending the epidemics of AIDS, TB, malaria and neglected tropical diseases, and combat hepatitis, water-borne diseases and other communicable diseases by 2030.

Infectious diseases are among top 10 causes of total number of deaths in the country dominated by diarrheal diseases, neonatal disorders, lower respiratory infections and tuberculosis. For India as a whole, the disease burden or DALY rate for diarrhoeal diseases, iron-deficiency anaemia, and tuberculosis is 2.5 to 3.5 times higher than the average globally.

Major Projects²

National HIV Cohort Programme

- The **National HIV Cohort Programme** is a collaborative effort jointly funded by the DBT and the International AIDS Vaccine Initiative (IAVI). This project's objective is to create well-characterized cohorts of individuals with HIV, establish a state-of-the-art biorepository for biological specimens, and develop a comprehensive national HIV/AIDS database.
- A network comprising six Centres of Excellence with ten locations throughout India was setup. Successfully achieved a total enrolment rate of 30%. To date, the department has collected and stored a total of 14,423 sample vials, including serum, plasma, and Peripheral blood mononuclear cells (PBMCs), in biorepositories. Furthermore, the programme has developed and implemented a stable version of the database at the National Institute of Epidemiology (NIE), Chennai, which is also operational at the various project sites.

National Liver Disease Biobank (NLDB), ILBS

- NLDB is a pioneering national initiative established through a collaborative effort between the **DBT** and **ILBS**. Its primary purpose is to support translational research in liver and related diseases. NLDB is certified by CTR.Net in Canada and accredited in accordance with ISO 20387:2018 standards. Furthermore, NLDB is playing a pivotal role in the development of three upcoming biobanks at **ICMR, All India Institute of Medical Sciences (AIIMS), and National Brain Research Centre (NBRC)**.
- Biobank provided 8,198 Next-generation sequencing (NGS), 6,510 Mass Spectrometry and 3,526 cell sorting and 8,200 animal pathology services. In 2022, NLDB also collected 25,000 samples and shared 8,772 bio samples.

Opportunities and Challenges

A. Opportunities



Facilitatory Ecosystem

Biotech parks and clusters created to foster sustainable growth in the research and development and innovation sector, a robust institutional mechanism is needed that actively promotes cross-sectoral research and supports innovation. Development like vaccine production, advanced diagnostics has enhanced public health laboratories and diagnostic centres, thereby improving the overall primary and secondary healthcare facilities.



Skilled Human Capital

India possesses a substantial reservoir of scientific human resources, comprising scientists and engineers.²⁶ ten Lakhs+ skilled biotech workers trained in Star College Mentorship Programme by DBT.²



Epidemiological factors

Increasing prevalence of various diseases is expected to drive the biotechnology market of India. Patient pool anticipated to surge over 20% in the next 10 years, mainly owing to rise in population. New diseases and lifestyle changes to accelerate demand for drugs and devices.



Research initiation in Rare Diseases

The Department of Biotechnology (DBT) has called for research and development (R&D) proposals on human genetic diseases centring on rare genetic disorders along with focusing on development of pharmacogenomic and other precision medicine tools to identify critical biomarkers for disease progression and drug responses in diverse populations.²⁷



Bio-informatics, genomics and precision medicine

200+ companies located in Bangalore, Delhi, Hyderabad, Pune and Chennai of bioinformatics are experiencing rapid growth within biotechnology sector of India. This expansion is being driven by both an increase in public funding for research²⁵ and development and a rise in investments from private companies.²⁶



Affordability

Rising income can drive 7 Crore households to the middle class over the next 10 years. Over 65 Crore people are expected to be covered by health insurance by 2020. The NHPS (National Health Protection Mission) is the largest govt. funded healthcare programme globally, anticipated to cover 10 Crore poor families in India by providing a benefit of up to INR 5 Lakhs per family per year.



The New Drugs and Clinical Trials Rules

The Ministry of Health and Family Welfare (MHFW) released new rule in 2019 to promote clinical research in India. Drug approval outcomes must now be shared within 30 days for clinical trials conducted in India, and within 90 days for clinical trials of drugs developed outside India. Previously, the approval process could take 6-12 months.



Advances in TB Research

Various studies, including BEAT (Building Evidence for Advance Treatment against Tuberculosis), aim to shorten XDR TB treatment to 6-9 months using Bedaquiline, Delamanid, Linezolid and Clofazimine. A C-TB skin test validation study is underway to detect TB infection. The SERB-DST supports NTEP with mathematical modelling and diverse TB research.²⁸

25. Kumar et. al (2011). *Biotechnology industry in India: Opportunities or challenges*

26. Invest India (2020). *Biotechnology, Science That Shapes Growth*

27. DBTIndia. (2023). *R&D Project Proposals*. Ministry of Science and Technology.

28. TBCIndia. (2022). *India TB Report 2022*.

B. Challenges



Human Resource

Ratio of researchers per million inhabitants of the country comes out to be 255 in 2017, which is low in comparison to Israel (8,342), Denmark (7,899) and Sweden (7,597).²⁸ Initiatives undertaken by DBT such as Science Setu Programme and Star College Mentorship Programme is anticipated to create career opportunities in biotechnology.¹



Clinical Trial Participation

Low participation in clinical trials, India's contribution to the global clinical trials has been ~1.5% as of February 2019.²⁹



Bio-networking

There is a requirement for bio-networking which is an effective tool for interaction and policy formulation in the biotechnology industry. DBT has allocated INR 523 Crores under the BIPP to support several programmes.³⁰ This will enable the industry to undertake high-risk, leading-edge research programmes.



Strategic roadmap

A strategic roadmap is needed to identify competitive areas and needs for industry-based R&D, and to make future plans taking into consideration the competencies and resources. DBT was allocated INR 2,581 Crores for 2022-23 which can help in developing strategic map defining major projects such as stem cell research, vaccine etc.²⁵

28. India Science and Technology (2019). *Research And Development Statistics at A Glance 2019-20*

29. BIRAC (n.d.). National Biopharma Mission - Oncology. Retrieved 19 September 2023

30. BIRAC (n.d.). *BIOTECHNOLOGY INDUSTRY PARTNERSHIP PROGRAMME (BIPP)*. Retrieved 19 September 2023

Cross-Cutting Themes



Climate

Significance: India faces environmental challenges due to climate change, impacting agriculture and health. Emphasizing sustainable biotech practices is vital to mitigate these effects and ensure a greener future.

Current efforts: The Indian government has launched the National Action Plan on Climate Change, Biotech-KISAN*, GOBARDhan* and many other schemes for promoting climate-resilient agriculture and sustainable energy solutions.

Way Forward: Collaborative efforts are essential, integrating renewable energy, circular economy approaches, and leveraging biotechnology for climate-smart agriculture to build a sustainable biotech industry.



Education

Significance: Education is a cornerstone for nurturing talent and fostering innovation in biotechnology. India's demographic advantage can be maximized by investing in quality education, skill development, and research capabilities.

Current efforts: Initiatives like DBT TWAS Fellowships, Khorana Programme for Scholars, B4 Programme* etc. and setting up Centres of Excellence in biotechnology education demonstrate the Indian government's commitment to enhancing biotech education and skills.

Way Forward: Increasing public-private partnerships, promoting industry-relevant education, and developing robust research ecosystems within academic institutions to elevate the biotech talent pool and foster innovation.



Research and Development

Significance: This forms the bedrock of biotech innovation, directly influencing economic growth. Robust research capacities can lead to breakthroughs in healthcare, agriculture, and industrial biotechnology.

Current efforts: India has significantly increased R&D investments in biotechnology, with DBT funding various research projects and promoting collaboration through initiatives like Global Bio-India amongst different biotechnology stakeholders.

Way Forward: Increasing public-private partnerships, promoting industry-relevant education, and developing robust research ecosystems within academic institutions to elevate the biotech talent pool and foster innovation.



Entrepreneurship

Significance: Entrepreneurship is crucial for fostering a diverse and vibrant biotech industry, particularly supporting startups and SMEs*. Accessible funding mechanisms can fuel innovation and entrepreneurship in this sector.

Current efforts: Government schemes like the Atal Innovation Mission and Make in India are promoting entrepreneurship and facilitating financial support to startups in biotech.

Way Forward: Simplifying regulatory frameworks, providing easier access to capital, and encouraging venture capital investments will promote financial inclusivity and nurture a dynamic biotech startup ecosystem.

*GOBARDhan: Galvanizing Organic Bio-Agro Resources Dhan

*Biotech-KISAN (Biotech Krishi Innovation Science Application Network)

*SME: Small and Medium Enterprises



Digital Technology

Significance: Integration of digital technology in the biotech sector, enable efficient processing, data analytics, and precision medicine, leading to significant advancements in healthcare and agriculture.

Current efforts: India's Digital India campaign and initiatives like the National Cold Chain and Management Information System, Niramai etc. are propelling the integration of digital technologies in biotechnology applications.

Way Forward: Continued investment in research and development of emerging technologies like AI, big data analytics, and fostering a culture of innovation to harness the potential of digitalization for the biotechnology industry.

Ministries and Departments

Inter-ministerial convergence among ministries and departments is particularly crucial in the context of disease programmes. Various ministries and departments work together to formulate policies and implement programmes aimed at preventing, controlling, and managing diseases. These collaborative efforts are essential to efficiently address public health challenges.



Ministry



Department and Bodies



Functions and Roles



विज्ञान एवं
प्रौद्योगिकी मंत्रालय
MINISTRY OF
SCIENCE AND
TECHNOLOGY

Ministry of Science and Technology



जैवप्रौद्योगिकी विभाग
DEPARTMENT OF
BIOTECHNOLOGY

Department of Biotechnology

It promotes and coordinates research, development, and application of biotechnology across various sectors. It plays a crucial role in fostering innovation and advancements in life sciences to address the nation's healthcare, agricultural, and environmental challenges individual well-being and family welfare, the department strives to ensure accessible and quality healthcare services for all citizens.



Biotechnology Industry Research Assistance Council (BIRAC)

A body that fosters and supports innovation, research, and entrepreneurship in the biotechnology sector. It plays a pivotal role in facilitating collaborations between academia, industry, and startups to accelerate the development and commercialization of biotechnological products.



स्वास्थ्य एवं
परिवार कल्याण मंत्रालय
MINISTRY OF
HEALTH AND
FAMILY WELFARE

Ministry of Health and Family Welfare



Central Drugs Standard Control Organization (CDSCO)

Responsible for drug standard control and grant of licenses for special category drugs.



Indian Council of Medical Research (ICMR)

Apex body in India for the formulation, coordination and promotion of biomedical research



रसायन एवं
उर्वरक मंत्रालय
MINISTRY OF
CHEMICALS
AND FERTILIZERS

Ministry of Chemicals and Fertilizers



Department of Pharmaceuticals
Ministry of Chemicals and Fertilizers
Government of India

Department of Pharmaceuticals

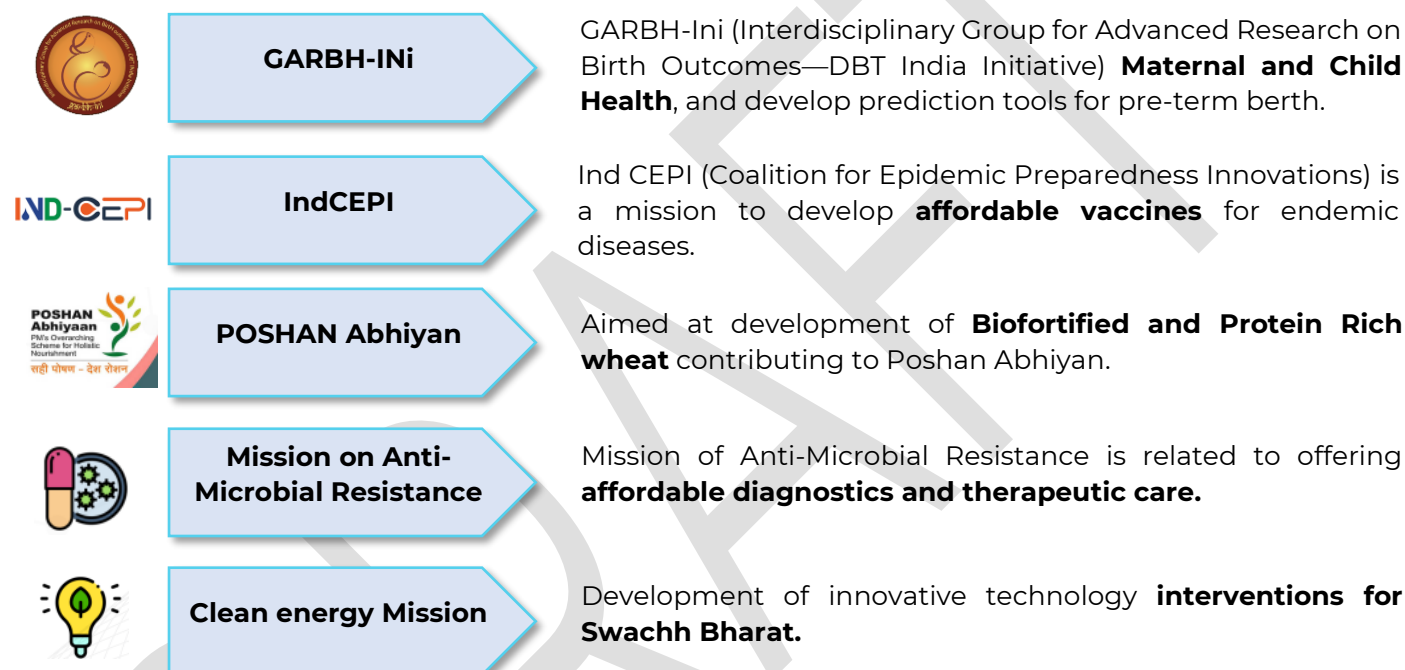
Nodal Department overseeing overall sector development and to regulate various complex issues related to pricing and availability of medicines

Government Interventions

As a response to the diverse health challenges existing within its population, India has developed a series of extensive policies and programmes designed to address a wide range of diseases. Various ministries, departments, and organisations work together to achieve common goals and objectives. These initiatives, encompassing various trends of biotechnology, shows the nation's dedication to enhancing healthcare access and outcomes throughout its expansive territory.

Prominent national-level programmes are briefly discussed below:

- 1. Atal Jai Anusandhan Biotech (UNaTI) Mission³¹** consists of various sub-missions like GARBH-ini for maternal and child health, Mission on Anti-Microbial Resistance, Ind CEPI(Coalition for Epidemic Preparedness Innovations) to develop infectious disease vaccines, Biofortification of food, and Clean Energy Mission for Swachh Bharat



- 2. National Biotechnology Parks Scheme³¹** establishes biotechnology parks in India to promote biotech product development, research, and entrepreneurship, by offering facilities for technology incubation, demonstration, and pilot plant studies, boosting the industry's growth.

1. Biotech Park, Lucknow, Uttar Pradesh:

- The Biotech Park has actively fostered the expansion of the bio-pharma, agri-biotechnology, and bio-service sectors by attracting companies to the state. It collaborates with state agencies and central research and development institutes and educational institutions to drive progress.
- Currently Biotech Park is holding hands with **four new entrepreneurs** namely M/s Helixion Biosciences Pvt. Ltd, M/s ABio Laboratories, M/s Grassroot Energy technologies India Pvt. Ltd, and M/s Green Vision Biotech with total 9 entrepreneurs.
- A total of **125 biotech/bioscience students** were trained in various areas of Biotechnology for a period of 1-6 months.

31. Department of Biotechnology (n.d.). Biotech Parks. Retrieved 19 September 2023

2. Biotechnology Incubation Centre, Hyderabad

- There are **11 incubatees**, graduated out from the park and operationalize.
- AACR (Advanced Analytical and Characterization Resource) facility established by The **Society for Biotechnology Incubation Centre** (SBTIC) and **Indian Institute of Chemical Technology** (IICT) with support from **BIRAC/DBT** has advanced instruments like, Multimode reader (UV-Vis, Fluorescence, Luminescence), Nuclear Magnetic Resonance (NMR), UHPLC-Q-TOF/MS, High-performance Liquid Chromatography (HPLC).

3. Technology Incubation Centre, Guwahati Biotech Park, Assam

- In the fiscal year 2021-22, the initiative provided laboratory space to nine entrepreneurs.
- A virtual internship programme focusing on Molecular Biology was successfully conducted.
- Additionally, GBP has entered into Memoranda of Understanding (MoUs) with notable organisations, including Social Alpha in Bangalore, Adamas University, KIIT-TBI, NEATeHUB, and Lumos Labs in Singapore.

4. KSUM-RGCB Innovation and Bioincubation Society (KRIBS)- Bio Nest, Kochi, Kerala

- The facility has 29 physical incubatees, nine virtual incubatees and 1 Young Innovators Programme (YIP) fellow.
- Some of the products developed by the incubatees such as **CoviKwik** (rapid immunochromatographic test for COVID-19 by Sushruta Innovations and Wellness), **Smartlyse hand sanitizer and wet wipes** by (Omics Ggen Life Sciences); **mobile phone sanitizer** (by Biophoton Technologies); **Coliform detection** kit (by Klones Life Sciences).

5. Industrial Biotechnology Parks (IBTPs) At Jammu and Kashmir

- The two Industrial Biotechnology Parks (IBTPs) (one IBTP at Kathua, Jammu and another at Kupwara, Kashmir Valley) were sanctioned by the DBT at a **total cost of Rs 8,466 Lakhs** for a period of three years.
- The establishment of these IBTPs accelerated bio-enterprise development along with employment generation for youth and students.

6. Chhattisgarh Biotech Park

- The park's objective is to establish a platform where research institutions can collaborate with industries to optimize the use of technology and available bioresources in New Raipur, located in the state of Chhattisgarh. This initiative is conducted in partnership with the Government of Chhattisgarh.

3. **National Biopharma Mission⁴** is a Pan-India programme with the main aim of making India a hub for design and development of novel, affordable and effective biopharmaceutical products and solutions.

The National Biopharma Mission (NBM), is a government-industry-academia collaboration dedicated to **Accelerating Discovery Research to Early Development for Biopharmaceuticals**, funded at a total cost of ₹1,500 Crores co-funded by the World Bank at 50% cost sharing. NBM's charter is being implemented by BIRAC, a Public Sector Undertaking, set up by the DBT. Its major focus is on development of vaccines, biosimilars, and medical devices.

Some objectives of this mission:³²

- In the next five years, the programme will help develop **6-10 new products**.
- To drive the growth curve for biopharma by accelerating the translational research concepts into viable products, enabling sustainable networks for collaboration between industry and academia.
- To bring isolated COE (centres of excellence), enhance regional capabilities and strengthen the current bio-clusters network.

32. BIRAC (n.d.). National Biopharma Mission. Retrieved 19 September 2023

4. Biotechnology Industry Partnership Programme (BIPP)³⁰

BIPP was launched in 2008. BIPP is a government partnership with Industries for support on research in frontier futuristic technology areas with economic potential and making the industry globally competitive.

BIPP supports the development of technologies in the context of recognized national priorities. Proposals are invited under 7 broad themes including drug delivery, vaccines and clinical trials, biosimilars and stem cells, devices and diagnostics, agriculture, bioinformatics.

Impact of BIPP

214 No. of projects supported	₹523 Crores Funds distributed	52 products/technologies developed	31 Intellectual property generated
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5. Star College Programme¹

The Star College Programme was initiated by DBT in 2008. Starting with 30 colleges in 2008, this scheme has supported more than 300 colleges all over the country.



Science Setu Programme

Programme is designed to build the significance of science. By engaging with professionals in the field and challenges of a career in science and also underscores the opportunities it presents.

Star College Mentorship Programme

The colleges are accredited with “Star Status” and adopted a higher role of mentoring the colleges that have currently initiated their journey under the “Strengthening Component.”

Significant outcomes of this programme

47 Rural colleges	11 Star status colleges	~3,000 Faculty beneficiaries	336 Lakhs Grant released (2022)
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6. Bio design Programme¹

In recognition of the significance of affordable, locally produced, and readily available medical technologies within the nation, the Bio design programme has been introduced as a leading initiative.

Its primary objective is to nurture and encourage the creation of affordable indigenous medical technologies while establishing an environment conducive to fostering innovation and entrepreneurship in the field of medical technology within the country.

Phase I:

4 bio design centres:

- School of International Bio design (SiB) centred at AIIMS and IIT Delhi
- Centre for Bio design and in-vitro Diagnostics (CBD) at Translational Health Science & Technology (THSTI), Faridabad
- Bio design and Bioengineering Initiative (BBI) programme at IISc., Bengaluru
- Healthcare Technology Innovation Centre (HTIC) at IIT Madras, Chennai

Phase II:

5 bio design centres:

- King George's Medical University (KGMU)
- IIT Bombay with Hinduja Hospital, Mumbai
- IIT-Jodhpur
- IIT-AIIMS
- THSTI, Faridabad with IIT Madras, CMC Vellore; PGIMER, Chandigarh and Venture Centre.

Significant outcomes of this programme

10	2	3	4
Research published	Patent filed	Process/product/Technology developed	Process/product/Technology commercialized

7. Biosafety Research Programme³³

Within the Biosafety Research programme, the primary focus is on facilitating the effective implementation of biosafety procedures, regulations, and guidelines mandated by the Environment (Protection) Act of 1986 and its corresponding Rules of 1989.

These measures are put in place to ensure the safety of both users and the environment when dealing with Genetically Modified Organisms (GMOs) and their derived products in research and practical applications.

8. Small Business Innovation Research Initiative (SBIRI) scheme¹

Supports development and initial validation of products/technologies up to Technology Readiness level 6 (TRL6).

³³. BIRAC (n.d.). Biosafety Programme. Retrieved 19 September 2023

Building Capacity



States need to ensure that biotechnology education at UG and PG levels is at par with the global standards to build a highly skilled workforce to cater to the biotech needs of the country. More focus needs to be put on creating state-of-the-art infrastructure facilities which can boost research, innovation, and development in the sector.¹

Promoting Biotech Startups



Infrastructure and Funding should be made available to build a strong startup ecosystem that can drive innovation and new technology in the sector. Development of affordable and accessible biotech products/technologies can be incentivised to further motivate entrepreneurs and startups.³⁴

One Nation One Health



Ind-Coalition of Epidemic Preparedness Innovations (CEPI) programme has been initiated by DBT to enable production of vaccine within 100 days. This initiative strengthens pandemic preparedness in keeping with the government's One Nation One Health aim.³⁵

Global Benchmarking



A framework for measuring the impact of biotechnology activities concerning global benchmarks and standards needs to be developed.

A data-monitoring unit can be set up to facilitate a data-based and objective assessment of the social and economic impact of research and development in the sector.

Building the Knowledge Base



To establish a robust biotechnology knowledge base, states need to offer incentives to attract and retain highly specialized workers, as knowledge workers currently constitute only 10.9% of total employment.³⁴

³⁴. BIRAC (n.d.). *Realising Investment Potential for Indian State-Specific Biotech Industries*. Retrieved 19 September 2023
³⁵. Indbiz (2023). *India to invest in developing any vaccine within 100 days*. Retrieved on 28 March 2023

Patent Expiration



India promotes research and manufacturing in biotechnology-based drugs with the help of Department of Biotechnology and BIRAC. Some of the initiatives for the boost includes policies to boost business development, guidelines for biologics regulatory pathway, capacity building, etc. Product patents of more than 10 blockbuster biologics with total revenue of INR 4 lakhs Crores are set to expire within next two to three years, providing a further boost to the biosimilars market, domestic companies could potentially benefit from the expiration of patents by developing biosimilars for the drugs that are set to expire which can help them gain foothold in the market and compete with established players.⁵

Increased FDI Flows



Foreign Direct Investment (FDI) in biotech is expected to have touched ₹6,906 Cr in 2021, up from ₹6,490 Cr in 2020. Some of major FDI commitments are by Mylan laboratories, Amneal Pharmaceuticals, Carlyle Group, KKR, Advent International. This is expected to be observed in the coming years.⁷

Research and Development



The industry is focusing on advancing in the competitive markets of generic and specialty medicine. The recently announced Centres of Excellence in the 2023-24 budget will foster pharmaceutical research and promote innovation.

Best Practices

1. Unique Methods of Management of Inherited Disorders (UMMID) Initiative- New Delhi, Jodhpur, Kolkata, Hyderabad (Major centres)²



Background:

The UMMID initiative, launched in 2019, aims to serve as a model framework for initiating network within the government healthcare system. UMMID operates across three closely interconnected tiers of healthcare, beginning with the establishment of diagnostic services, followed by training clinicians in the diagnosis and management of genetic diseases, and finally, conducting outreach activities to extend these services to vulnerable segments of the population.

The screening programme primarily targets pregnant women to identify hemoglobinopathies like thalassemia and sickle cell disease, as well as newborns for five major genetic disorders that require early intervention. These disorders include congenital hypothyroidism, congenital adrenal hyperplasia, galactosemia, and G6PD deficiency.



Steps taken by the Government

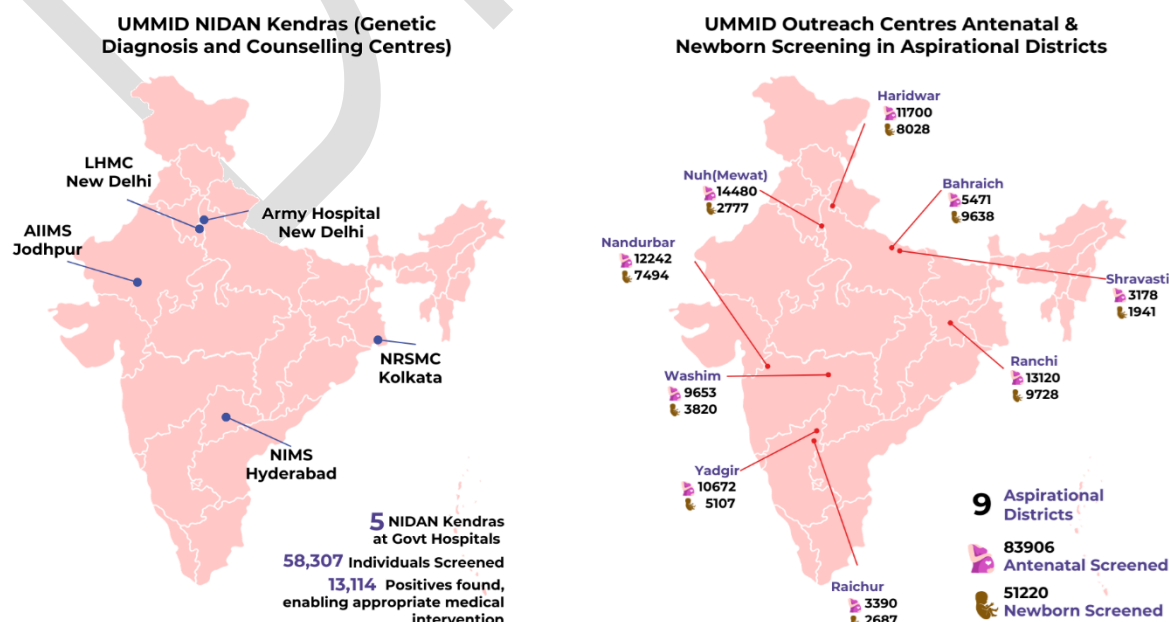
The Department has provided assistance in setting up 5 NIDAN Centres (genetic testing laboratories), along with 6 Training Centres and 7 Outreach Centres for the screening of pregnant women and neonates across 9 Aspirational Districts



Impact generated

- The five NIDAN Centres have offered screening services to over 58,000 beneficiaries to date, and have successfully identified genetic abnormalities in over 13,000 cases.
- The seven Outreach Centres have conducted genetic screenings for over 83,900 pregnant women and 51,200 newborns across nine Aspirational Districts. This effort has contributed to averting approximately 34 births of children affected by hemoglobinopathies and has identified over 150 carrier couples with a 1 in 4 risk of having an affected child in the future. Additionally, more than 350 newborns received timely diagnoses, facilitating early treatment.
- The six Training Centres have provided comprehensive, practical training programmes lasting up to six months in genetic diagnosis and counselling. These programmes have benefited 47 government doctors hailing from 18 different states and union territories, representing all six geographical zones.

Figure 4 - UMMID NIDAN Kendras and outreach centres



2. Biomedical Engineering Research Programmes- Hyderabad (Telangana), Surat (Gujarat)²



Background

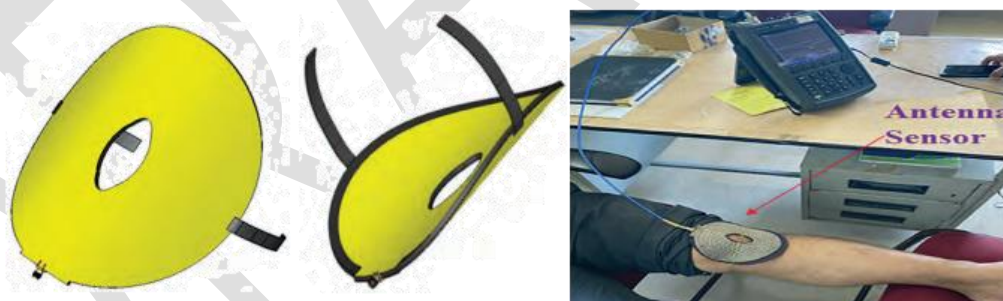
Biomedical engineering is an interdisciplinary area of research involving engineering and other quantitative sciences for unravelling the complexities of biological systems and provides cost-effective solutions for improved quality of life. The division of biomedical engineering fosters and supports innovative ideas in the field of bio-medical devices, diagnostics, implants bioinstrumentation, tissue engineering for development of affordable healthcare technologies and enhancing their access.



Steps taken by the Government

- **Development of technology for manufacturing microneedles (MNs) on small scale:** A project being implemented at Dr. Reddy's Institute of Life Sciences- Hyderabad, has led to development of technology for manufacturing microneedles (MNs) on small scale. Comparative study of the drug release from two types of the formulation (drug was loaded either as amorphous powder, or in form of nanoparticles) demonstrated advantage of the NPs containing formulation. This is being explored for Iron and Vitamin B-12 supplementation in women.
- **Development of wearable textile antenna sensor device for detection of knee effusion:** In a project implemented at Sardar Vallabhbhai National Institute of Technology, Surat, a compact, lightweight, robust and stable wearable textile antenna sensor device has been designed for detection of knee effusion. The textile antenna is fabricated using laser cutting process and embroidery technique employing silver coated conductive thread. The prototype design was successfully tested on knee aspiration anatomy model for knee effusion detection and has shown good agreement with the simulated results

Figure 5 - Textile Antenna based wearable bio sensor device for detection of knee effusion



3. Inauguration of Council Of Scientific And Industrial Research–Indian Institute Of Integrative Medicine's (CSIR-IIIM) BioNEST- Bio incubation Centre in Jammu³⁶



Background:

The UMMID initiative, launched in 2019, aims to serve as a model framework for initiating network within the government healthcare system. UMMID operates across three closely interconnected tiers of healthcare, beginning with the establishment of diagnostic services, followed by training clinicians in the diagnosis and management of genetic diseases, and finally, conducting outreach activities to extend these services to vulnerable segments of the population.

The screening programme primarily targets pregnant women to identify hemoglobinopathies like thalassemia and sickle cell disease, as well as newborns for five major genetic disorders that require early intervention. These disorders include congenital hypothyroidism, congenital adrenal hyperplasia, galactosemia, and G6PD deficiency.

³⁶. Indian Institute of Integrative Medicine (IIIM) (n.d.). *Inauguration Ceremony of BioNEST Bio incubation Centre has been graciously inaugurated by Dr. Jitendra Singh, Hon'ble Union Minister at CSIR-IIIM, Jammu.* Retrieved 19 September 2023



Steps taken by the Government

CSIR-IIIM, Jammu's BioNEST- Bio-incubator was inaugurated by Union Minister of State (Independent Charge) S&T, Dr Jitendra Singh, on 23 April 2022. The objectives of this incubator are to ignite the entrepreneurship mindset and to nurture start-up culture among youth, local farmers and entrepreneurs of Jammu & Kashmir.

Bio-NEST incubation centre has been financially supported by DBT-BIRAC to foster the biotech innovation ecosystem in the country. Bio-NEST programme provides support to establish bio-incubators either as a standalone entity or as a part of academia.



Impact generated

- 64 Start-Ups have already registered with CSIR-IIIM Jammu and a fresh impetus has been given to promote Start-Up as an alternative source of livelihood, with financial, technical and logistic support being provided by Union Ministry of Science and Technology through its different agencies and departments. Out of the 64 Startups, 14 have developed products and 4 have already reached the market.

4. Virtual Network Centres in Cancer Immunology and Immunotherapy³⁷



Background:

Cancer is a major public health concern in India. India sees an incidence of more than 1 million new cases of cancer every year, a number that is likely to increase given the increasing age of Indian population and lifestyle changes.

Department of Biotechnology (DBT) has a clear definitive role in the management and control of cancer. Through Research & Development programmes, multi-institutional networking programmes, DBT has provided resources to individual investigators and to institutions, to carry out basic, applied, translational and clinical research and to develop new methods to prevent and treat disease and to conduct research especially in challenging areas pertaining to different types of cancer.



Steps taken by the Government

The Department has supported various projects and initiatives including International Cancer Genome Consortium (ICGC) DBT India Project, Systems Medicine Cluster (SyMeC), Kalyani Cluster, Virtual Network Centres in Cancer Immunology and Immunotherapy, DBT-AIIMS-NCI India Translational and Clinical Research Partnership Centre, and DBT DAE NCG Partnership.

- **International Cancer Genome Consortium (ICGC) DBT India Project:** ICGC-India Project has established a dedicated bio-repository with best international practices at ACTREC, Mumbai and State-of-the-art high throughput Genome sequencing centre at NIBMG. The research has led to the identification of certain signatures in oral cancer as biomarkers for early detection.
- **Systems Medicine Cluster (SyMeC), Kalyani Cluster:** SyMeC's research has already resulted in the development of a powerful statistical and bioinformatics method to integrate genotype and gene-expression data for dissecting the genetic architecture of a disease. Further, since lymph node (LN) metastasis is the most important prognostic factor in oral cancer with many (30-50%) oral cancer patients developing lymph node metastasis, the investigators have carried out systematic genome-scale investigations. They have identified that lymph node metastasis in oral cancer is driven by chromosomal

³⁷. Department of Biotechnology (n.d.). *Human Genomics Oncology- Cancer*. Retrieved 19 September 2023

- **Virtual Network Centres in Cancer Immunology and Immunotherapy:** Three VNCs have been recommended for support and implementation; (a) A Phase 2 Randomized Controlled Clinical Trial to Evaluate the Role of Metronomic Chemotherapy and Dendritic Cell Vaccine in Recurrent Hormone Receptor Negative Breast Cancer at Adyar Cancer Institute, Chennai; (2) Development of Genetically Engineered 'Off-the-shelf' and Inducible CAR-T Cells for Cancer Therapeutics at IGIB, New Delhi, and (3) Network Centre for Research on Glioblastoma Stem cell-targeted T-Cell Immunotherapy using Non-Genetically Engineered Mesenchymal Stromal Cells at Siddaganga Institute of Technology, Karnataka.
- **DBT-AIIMS-NCI India Translational & Clinical Research Partnership Centre:** DBT has recently recommended for the establishment of DBT-AIIMS-NCI India Translational & Clinical Research Partnership Centre at NCI Jajjar as 'National Cancer Centre for India'. The partnership centre envisages to (1) Establish a state-of-the-art Cancer Genomic Research Facility for Personalized Medicine; (2) Establish Early Detection Research Network (EDRN) Programme with International Partnership; (3) Establish a quality enabled "Bio-bank" for "India Centric Cancers" enhancing exploration and validation of novel diagnostic or therapeutic modalities, and (4) Establish an advanced animal facility for gene-manipulated and immunocompromised small animals for preclinical cancer research and technology validation needed for planning of clinical trials.



Impact generated

- DBT has supported a variety of team science approaches. DBT identified Virtual National Cancer Institutes (VNCIs), one for Hormone Resistant Breast Cancer and the other for Oral Cancer. VNCI Breast has been successful in establishing Patient Derived Xenograft (PDX) Models for drug screening for first time in India.
- DBT implemented a programme entitled 'Pilot Projects for Young Investigators' programme and witnessed 166 publications.
- Cancer Diagnostics Centre, SKIMS, Kashmir has been providing high end molecular biology based diagnostic investigations to the patients of J&K in general and to the patients of Kashmir valley.

Recommended Readings

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